Exploratory Data Analysis (EDA) Assignment using Pandas

Objective:

Learn basic pandas operations, distribution analysis, skewness detection, normalization, standardization, transformations, and visualization using pairplot and boxplot.

Section A: Basic Pandas Operations (10 Marks)

Load dataset using pandas.

Display first 5 rows, last 5 rows, shape, and column names.

Display statistical summary using describe().

Check for missing values.

Section B: Distribution Analysis (10 Marks)

Plot histogram of any 3 numerical columns.

Plot distribution using seaborn.

Explain whether data is normally distributed or skewed.

Section C: Skewness Analysis (10 Marks)

Calculate skewness of all numerical columns.

Identify which columns are positively and negatively skewed.

Section D: Transformation (15 Marks)

Apply log transformation.

Apply square root transformation.

Apply square transformation.

Apply cube transformation.

Compare skewness before and after transformation.

Section E: Normalization and Standardization (15 Marks)

Perform normalization using Min-Max scaling.

Perform standardization using Z-score.

Compare mean and standard deviation before and after standardization.

Section F: Visualization (20 Marks)

Create boxplot for 3 numerical columns.

Identify outliers.

Create pairplot.

Explain relationships between variables.

Section G: Interpretation (20 Marks)

Which column has highest skewness?

Which transformation worked best?

Did normalization change distribution?

What insights did you get from pairplot?

Bonus Question (Optional - 10 Marks)

Create correlation heatmap and explain correlation.